How to execute Assignment 1?

Step-by-Step Execution of Hadoop MapReduce IP Count Assignment (From Scratch)

Step 1 — Open Eclipse

Start Eclipse IDE.

Step 2 — Create New Java Project

Go to:

```
File → New → Java Project
```

Project Name:

```
map_reduce
```

Click:

```
Finish
```

Step 3 — Create Package

Inside project:

```
src → Right Click → New → Package
```

Package Name:

```
map_reduce
```

Click:

```
Finish
```

Step 4 — Create Java Class

Right click package:

```
New → Class
```

Class Name:

```
map_reduce
```

Click:

```
Finish
```

Step 5 — Write MapReduce Program

Write Hadoop MapReduce code in:

```
map_reduce.java
```

Program functionality:

- Mapper extracts IP address

- Reducer counts occurrences

Step 6 — Add Hadoop JAR Files

Right click project:

```
Build Path → Configure Build Path
```

Open:

```
Libraries
```

Click:

```
Add External JARs
```

Step 7 — Add JARs from Hadoop Folders

Add ALL `.jar` files from:

common

```
C:\hadoop\share\hadoop\common
```

---

common/lib

```
C:\hadoop\share\hadoop\common\lib
```

---

mapreduce

```
C:\hadoop\share\hadoop\mapreduce
```

---

hdfs

```
C:\hadoop\share\hadoop\hdfs
```

---

yarn

```
C:\hadoop\share\hadoop\yarn
```

Inside every folder:

```
Ctrl + A → Open
```

Step 8 — Fix Java Version Compatibility

Right click project:

```
Properties
```

Open:

```
Java Build Path → Libraries
```

Remove:

```
JavaSE-17
```

Add:

```
Add Libraries → JRE System Library → jre
```

Click:

```
Finish
```

Step 9 — Change Compiler Level

Open:

```
Java Compiler
```

Uncheck:

```
Use compliance from execution environment JavaSE-17
```

Set:

```
Compiler compliance level = 11
```

Click:

```
Apply and Close
```

Step 10 — Clean Project

Top Menu:

```
Project → Clean
```

Select:

```
Clean all projects
```

Click:

```
Clean
```

Step 11 — Export JAR File

Right click project:

```
Export
```

Select:

```
Java → JAR file
```

Click:

```
Next
```

Select project files.

Save JAR as:

```
map_reduce.jar
```

Location:

```
Downloads folder
```

Click:

```
Finish
```

Step 12 — Open Ubuntu WSL Terminal

Start Ubuntu WSL terminal.

Step 13 — Start Hadoop DFS

Run:

```
start-dfs.sh
```

Step 14 — Start Hadoop YARN

Run:

```
start-yarn.sh
```

Step 15 — Verify Hadoop Services

Run:

```
jps
```

Expected processes:

```
NameNode
DataNode
SecondaryNameNode
ResourceManager
NodeManager
```

Step 16 — Keep Input File Ready

Input file:

```
log.txt
```

Contains log entries with IP addresses.

Example:

```
10.1.1.236 user login
10.1.1.236 user logout
10.10.55.142 user login
```

Step 17 — Run Hadoop MapReduce Program

Run command:

```
hadoop jar /mnt/c/Users/54642/Downloads/map_reduce.jar map_reduce.map_reduce log.txt output_new2
```

Step 18 — Hadoop Processing Begins

Internally:

- Mapper reads input

- Extracts first token(IP)
- Emits `(IP,1)`
- Shuffle and Sort groups same IPs
- Reducer sums counts

Step 19 — Successful Execution

Observe:

```
map 100%
reduce 100%
Job completed successfully
```

Step 20 — Display Output

Run:

```
hadoop fs -cat output_new2/part-r-00000
```

Step 21 — Final Output

Output format:

```
IP_Address    Count
```

Example:

```
10.103.190.81   53
10.82.30.199    63
```

Meaning:

- IP `10.103.190.81` appeared 53 times
- IP `10.82.30.199` appeared 63 times

Working Principle Summary

Mapper

Extracts IP address and emits:

```
(IP,1)
```

Reducer

Counts total occurrences of each IP.

Final Conclusion

The Hadoop MapReduce program successfully analyzed the log file and counted frequency of each IP address using Mapper and Reducer in Hadoop framework.